

PQC-Readiness Audit Report

Contract: /mnt/d/Crypto/audit_case_studies/contracts/GnosisSafe_Safe.sol
SHA-256: 9f91d9250e18bb0710b7b1f10dcbe241... **Lines:** 576 **Date:** 2026-08-20 **Tool:** pqc-audit v0.2.0

Executive Summary

Check	Result	Detail
Security Scan	WARN	0 critical, 5 high
PQC Readiness	FAIL	2 quantum-vulnerable primitives, 0 critical implementation advisories
Gas Optimization	15 opportunities	~4205 gas units
OVERALL	FAIL	

Phase 1: Structure

- **Solidity Version:** $\geq 0.7.0$ $< 0.9.0$
- **Contracts:** 17
- **Functions:** 16
- **Events:** 1

Phase 2: Gas Optimization (15 found)

#	Pattern	Impact	Fixable
1	Use calldata instead of memory for read-only function parameters	Info	Manual
2	Use calldata instead of memory for read-only function parameters	Info	Manual
3	Use calldata instead of memory for read-only function parameters	Info	Manual

#	Pattern	Impact	Fixable
4	Use != 0 instead of > 0 for unsigned integers	Info	Manual
5	Use != 0 instead of > 0 for unsigned integers	Info	Manual

Phase 3: Security Analysis (16 findings)

Severity	Count
Critical	0
High	5
Medium	11
Low	0

Security Score: 0.0/100

Top Findings

- **[HIGH]** Line 253 — Unchecked external call
 - Code: `(bool refundSuccess, ) = receiver.call{value: payment}("");`
 - External call return value not checked. Always check return values from `send()`, `call()` and external calls.
- **[HIGH]** Line 217 — Use of `tx.origin` for authorization
 - Code: `// We transfer the calculated transaction costs to the tx.origin to avoid sending it to intermediate contracts that have made calls.`
 - Using `tx.origin` for authorization makes a contract vulnerable to phishing attacks.
- **[HIGH]** Line 238 — Use of `tx.origin` for authorization
 - Code: `* @param refundReceiver Address of receiver of the gas payment (or 0 for tx.origin).`
 - Using `tx.origin` for authorization makes a contract vulnerable to phishing attacks.
- **[HIGH]** Line 249 — Use of `tx.origin` for authorization
 - Code: `address payable receiver = refundReceiver == address(0) ? payable(tx.origin) : refundReceiver;`
 - Using `tx.origin` for authorization makes a contract vulnerable to phishing attacks.

- **[HIGH]** Line 561 — Use of tx.origin for authorization
 - Code: * @param refundReceiver Address of receiver of gas payment (or 0 fortx.origin).
 - Using tx.origin for authorization makes a contract vulnerable to phishing attacks.

Phase 4: Post-Quantum Cryptography Readiness

Line	Primitive	Risk	Migration Path
400	ecrecover	CRITICAL	Replace with ML-DSA-65 (Dilithium-3) via a FIPS 204 implemen. . .
62	keccak256	MEDIUM	Double hash output length or use SHA-3-512. . .
420	ecdsa	CRITICAL	ML-DSA-65 (FIPS 204). . .

Quantum Risk Level: CRITICAL

NIST Status: FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) finalized. FIPS 206 (FN-DSA/Falcon) selected for standardization; publication pending.

Implementation Vulnerability Scan (2024-2026 Research)

Vulnerability	Risk	Source
<i>None detected</i>	LOW	No known implementation vulnerabilities match this contract

Evidence Attestation

- **Source SHA-256:** 9f91d9250e18bb0710b7b1f10dcbe2417e9ef8d22eeee98dba2ef48661a060fa
- **Report SHA-256:** dcbf8a695e89e27953c138d630fc984a. . .

This report was generated mechanically. No LLM inference was used in the security or PQC assessments. PQC-Readiness Audit — pqc-audit v0.2.0